Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming 1st mid-term test– November 12th 2024

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:15m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and ensure to provide a brief explanation. Remember, without an explanation, the answer will be considered invalid. Understanding the questions is crucial for your success in this exam.

A. "In a PRAM model, synchronization primitives are not required when data is exchanged through the shared memory." True/False? Why? (1)

B. Please show the formula defining the "Efficiency" in a PRAM model. (1)

C. Can you provide an example where the "work" done by a parallel PRAM program exceeds that of the sequential version of the same algorithm? (1)

D. What does it mean fixed-size model? Does it refer to Gustafson or Amdahl's law? (1)

Exercise n. 2

Answer the following questions about advanced algorithms algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the ideas used to improve the Karger approach to solving the min-cut problem. What changed with respect to the asymptotic complexity with respect to the initial approach? (2)

B. Master Theorem method is an important tool for the Divide and Conquer class of algorithms. Why and under which assumptions the theorem could be applied? (1)

C. What is meant by the memoization technique, and in which class of algorithm does it matter? (1)

Exercise n. 3

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the differences between Las Vegas and Monte Carlo algorithms, even considering time complexity issues. (2)

B. Describe how the split for Treaps data structure works and discuss its complexity. (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Please explain what "Lower bound for comparison sorting" means and its implications. (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

B. Please show the pseudo-code of the partitioning algorithm used by Quicksort and report its variants, complexity, and usage in other algorithms. (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**

C. Why is the Counting sort used as the auxiliary algorithm of the Radix sort? (1)